Assignment 2 - Momentum strategies

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Question a

We construct a cross-sectional long–short industry momentum strategy using the 30 Industry Portfolios from Ken French's data library (monthly total returns). Our sample period spans June 1964 to August 2025, providing more than 60 years of monthly data.
Each month, industries are ranked according to their past 12–2 month cumulative return, which are the returns from months $t-12$ to $t-2$, excluding the most recent month to avoid short-term reversal effects.
We then:

- **Go long** the top 20% of industries (strongest momentum),
- **Go short** the bottom 20% (weakest momentum),
- Use **equal weights** within each leg.

The monthly strategy return is computed as:

$$r_t = R_t^{\text{long}} - R_t^{\text{short}}$$

and represents an excess return relative to the short leg.
Starting from \$1 at the beginning of the sample, the cumulative value is updated as:

$$V_t = V_{t-1} \times (1 + r_t)$$

The geometric mean monthly return of the raw strategy is **0.63%**, equivalent to approximately **7.85% per year**. The arithmetic mean monthly return is **0.74%** (or about **9.28% annualized**).
The portfolio exhibits strong long-run gains but occasional large draw-downs, particularly during market regime reversals. Here is the evolution of the portfolio across time :

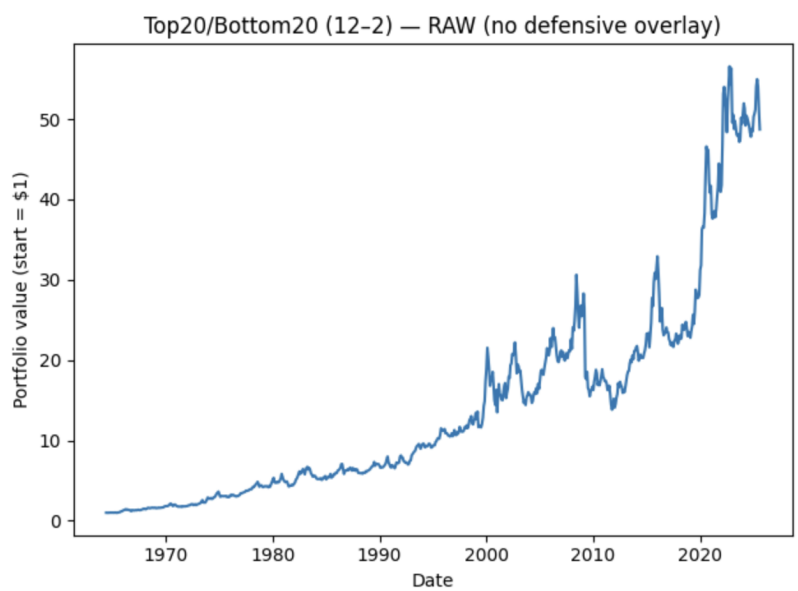

Figure 1: Portfolio with top20/bottom20 raw strategy

Question b

To improve risk-adjusted performance, we design a **defensive version** of the strategy incorporating volatility targeting and regime-based scaling. This overlay aims to mitigate deep drawdowns without significantly reducing long-term returns.

Volatility Targeting

Each month, we estimate a 12-month rolling realized volatility of the raw long–short returns (lagged one month). We then scale the exposure to target a monthly volatility of 4%:

$$\text{Leverage}_t = \min\left(\frac{0.04}{\hat{\sigma}_{t-1}}, 2.0\right)$$

where the leverage is capped at 2.0x.

Bear-Market Clamp

When the lagged 12-month market total return (Mkt + RF) is negative, exposure is reduced by 30%:

$$\text{Leverage}_t = 0.7 \times \text{Leverage}_t$$

Results

The defensive adjustment substantially reduces drawdowns. The maximum draw-down decreases from roughly **55%** (raw) to **27%** (defensive). Average annual return remains near **10%**, indicating that the overlay enhances stability rather than sacrificing performance.
Here is the evolution of the portfolio of this defensive strategy across time :

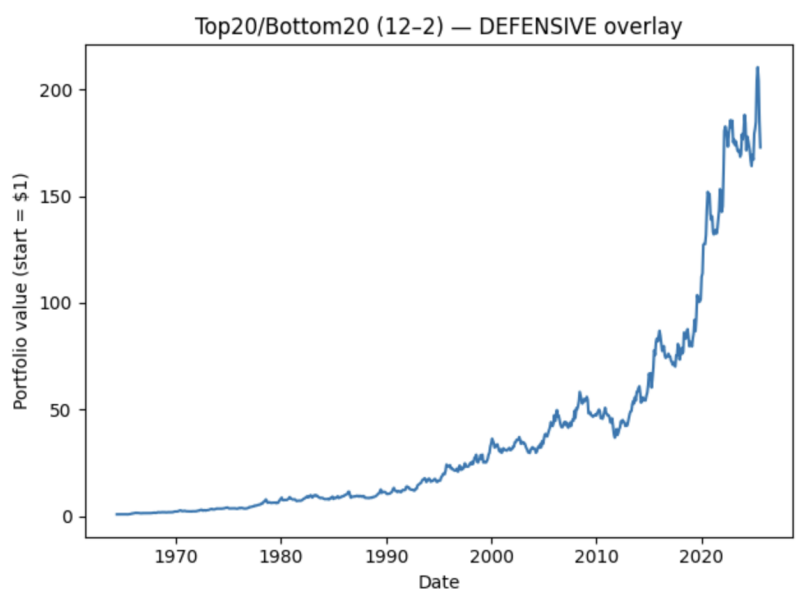

Figure 2: Portfolio with top20/bottom20 defensive strategy

Question c

We evaluate both strategies relative to the **Fama–French 3-factor (FF3)** and **Fama–French 5-factor (FF5)** models:

$$r_t = \alpha + \beta_M (Mkt - RF)_t + \beta_S SMB_t + \beta_H HML_t + \varepsilon_t$$

$$r_t = \alpha + \beta_M (Mkt - RF)_t + \beta_S SMB_t + \beta_H HML_t + \beta_R RMW_t + \beta_C CMA_t + \varepsilon_t$$

Monthly factor returns are sourced from Ken French's data library. The estimated alphas (intercepts) correspond to the portion of returns unexplained by standard risk factors.

Model	**Strategy**	**Monthly α**	**Annualized α**
FF3	Raw	0.800%	10.04%
FF3	Defensive	0.888%	11.20%
FF5	Raw	0.725%	9.05%
FF5	Defensive	0.857%	10.79%

Both the raw and defensive versions display **positive alphas**. The defensive overlay slightly increases the estimated alpha in both the FF3 and FF5 frameworks, from 0.80% to 0.89% per month under FF3 and from 0.73% to 0.86% under FF5. This improvement suggests that the volatility scaling and market-regime adjustment enhance risk-adjusted performance without sacrificing excess return. The persistence of significant alphas indicates that cross-industry momentum is not fully explained by traditional risk factors.
Here are the full results of the linear regressions :

```
================ FF3 Raw Strategy ================

                            OLS Regression Results
==============================================================================
Dep. Variable:                ret_raw   R-squared:                       0.038
Model:                            OLS   Adj. R-squared:                  0.034
Method:                 Least Squares   F-statistic:                     9.851
Date:                Sun, 12 Oct 2025   Prob (F-statistic):           2.24e-06
Time:                        22:21:54   Log-Likelihood:                 1256.4
No. Observations:                 746   AIC:                            -2505.
Df Residuals:                     742   BIC:                            -2486.
Df Model:                           3
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0080      0.002      4.775      0.000       0.005       0.011
Mkt-RF        -0.1424      0.039     -3.619      0.000      -0.220      -0.065
SMB            0.0243      0.057      0.428      0.668      -0.087       0.136
HML           -0.2668      0.057     -4.692      0.000      -0.378      -0.155
==============================================================================
Omnibus:                       94.817   Durbin-Watson:                   1.913
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              415.739
Skew:                          -0.494   Prob(JB):                     5.29e-91
Kurtosis:                       6.521   Cond. No.                         37.1
==============================================================================
```

```
================ FF3 Defensive Strategy ================

                            OLS Regression Results
==============================================================================
Dep. Variable:                ret_def   R-squared:                       0.022
Model:                            OLS   Adj. R-squared:                  0.018
Method:                 Least Squares   F-statistic:                     5.652
Date:                Sun, 12 Oct 2025   Prob (F-statistic):           0.000785
Time:                        22:21:54   Log-Likelihood:                 1304.8
No. Observations:                 746   AIC:                            -2602.
Df Residuals:                     742   BIC:                            -2583.
Df Model:                           3
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0089      0.002      5.655      0.000       0.006       0.012
Mkt-RF        -0.0456      0.037     -1.237      0.217      -0.118       0.027
SMB            0.0583      0.053      1.096      0.273      -0.046       0.163
HML           -0.2149      0.053     -4.034      0.000      -0.320      -0.110
==============================================================================
Omnibus:                       13.833   Durbin-Watson:                   1.923
Prob(Omnibus):                  0.001   Jarque-Bera (JB):               23.547
Skew:                          -0.083   Prob(JB):                     7.71e-06
Kurtosis:                       3.854   Cond. No.                         37.1
==============================================================================
```

```
================ FF5 Raw Strategy ================

                            OLS Regression Results
==============================================================================
Dep. Variable:                ret_raw   R-squared:                       0.049
Model:                            OLS   Adj. R-squared:                  0.043
Method:                 Least Squares   F-statistic:                     7.671
Date:                Sun, 12 Oct 2025   Prob (F-statistic):           4.83e-07
Time:                        22:21:54   Log-Likelihood:                 1260.7
No. Observations:                 746   AIC:                            -2509.
Df Residuals:                     740   BIC:                            -2482.
Df Model:                           5
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0072      0.002      4.241      0.000       0.004       0.011
Mkt-RF        -0.1077      0.041     -2.620      0.009      -0.188      -0.027
SMB            0.0413      0.060      0.690      0.490      -0.076       0.159
HML           -0.4187      0.077     -5.438      0.000      -0.570      -0.268
RMW            0.0400      0.081      0.495      0.621      -0.119       0.199
CMA            0.3386      0.116      2.922      0.004       0.111       0.566
==============================================================================
Omnibus:                       93.428   Durbin-Watson:                   1.916
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              393.291
Skew:                          -0.499   Prob(JB):                     3.96e-86
Kurtosis:                       6.414   Cond. No.                         80.2
==============================================================================


================ FF5 Defensive Strategy ================

                            OLS Regression Results
==============================================================================
Dep. Variable:                ret_def   R-squared:                       0.024
Model:                            OLS   Adj. R-squared:                  0.017
Method:                 Least Squares   F-statistic:                     3.579
Date:                Sun, 12 Oct 2025   Prob (F-statistic):            0.00332
Time:                        22:21:54   Log-Likelihood:                 1305.3
No. Observations:                 746   AIC:                            -2599.
Df Residuals:                     740   BIC:                            -2571.
Df Model:                           5
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0086      0.002      5.327      0.000       0.005       0.012
Mkt-RF        -0.0343      0.039     -0.885      0.376      -0.110       0.042
SMB            0.0697      0.056      1.235      0.217      -0.041       0.180
HML           -0.2611      0.073     -3.601      0.000      -0.404      -0.119
RMW            0.0386      0.076      0.507      0.612      -0.111       0.188
CMA            0.0997      0.109      0.914      0.361      -0.115       0.314
==============================================================================
Omnibus:                       13.230   Durbin-Watson:                   1.921
Prob(Omnibus):                  0.001   Jarque-Bera (JB):               22.458
Skew:                          -0.069   Prob(JB):                     1.33e-05
Kurtosis:                       3.839   Cond. No.                         80.2
==============================================================================
```

Python code

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import statsmodels.api as sm


### Inputs
IND_PATH = "30_Industry_Portfolios.csv"
FF3_PATH = "F-F_Research_Data_Factors.csv"
FF5_PATH = "F-F_Research_Data_5_Factors_2x3.csv"

ind = pd.read_csv(IND_PATH, sep=";")
ff3 = pd.read_csv(FF3_PATH, sep=";")
ff5 = pd.read_csv(FF5_PATH, sep=";")

def parse_yyyymm(s):
    s = s.astype(str).str.strip()
    dt = pd.to_datetime(s, format="%Y%m", errors="coerce")
    return dt.fillna(pd.to_datetime(s, format="%Y-%m", errors="coerce"))

# Normalize date columns
ind["DATE"] = parse_yyyymm(ind.pop("Date"))
ff3["DATE"] = parse_yyyymm(ff3.pop("Date"))
ff5["DATE"] = parse_yyyymm(ff5.pop("Date"))

# Convert % to decimals
def to_decimal(df, exclude=("DATE",)):
    cols = [c for c in df.columns if c not in exclude]
    for c in cols:
        df[c] = pd.to_numeric(df[c], errors="coerce")
    if any(df[c].abs().max(skipna=True) > 1.0 for c in cols):
        for c in cols:
            df[c] = df[c] / 100.0
    return df

ind = to_decimal(ind)
ff3 = to_decimal(ff3)
ff5 = to_decimal(ff5)

# Use FF5 factors
factors = ff5[["DATE","Mkt-RF","SMB","HML","RMW","CMA","RF"]].copy()




### Strategy 1 : 12-2 momentum, Top20/Bottom20
lookback_start, lookback_end = 2, 12
frac = 0.20
industry_cols = [c for c in ind.columns if c != "DATE"]
n = len(ind)

# Compute 12-2 momentum
mom_vals = []
for t in range(n):
    start, end = t - lookback_end, t - lookback_start
    if start < 0:
        mom_vals.append(pd.Series(index=industry_cols, dtype=float))
    else:
        window = ind.iloc[start:end+1][industry_cols]
        mom_vals.append((1.0 + window).prod() - 1.0)
mom = pd.DataFrame(mom_vals)
mom.insert(0, "DATE", ind["DATE"].values)

# Long-short (Top 20% - Bottom 20%) excess return
ls_rows = []
for t in range(n):
    row = mom.iloc[t][industry_cols]
    if row.isna().all():
        continue
    ranks = row.sort_values(ascending=False)
    k = max(1, int(round(frac * len(ranks))))
    long_names, short_names = ranks.index[:k], ranks.index[-k:]
    long_r = ind.iloc[t][long_names].mean()
    short_r = ind.iloc[t][short_names].mean()
    ls_rows.append({"DATE": mom.iloc[t]["DATE"], "ret_raw": long_r - short_r})
ls = pd.DataFrame(ls_rows)
```

```
### Strategy 2 : Defensive overlay
target_vol, cap_lev, bad_mkt_mult = 0.04, 2.0, 0.7

# Merge minimal factors for overlay math
ls = ls.merge(factors[["DATE","Mkt-RF","RF"]], on="DATE", how="left")

# 12-month realized vol of raw returns (lag 1 month)
ls["raw_vol_12m"] = ls["ret_raw"].rolling(12).std().shift(1)
ls["lev"] = (target_vol / ls["raw_vol_12m"]).clip(upper=cap_lev)

# Reduce exposure if trailing 12m market total return is negative
tmp = factors.copy()
tmp["MKT_TOTAL"] = tmp["Mkt-RF"] + tmp["RF"]
mkt_total_12m = (1.0 + tmp["MKT_TOTAL"]).rolling(12).apply(np.prod, raw=False) - 1.0
ls = ls.merge(pd.DataFrame({"DATE": tmp["DATE"], "mkt_12m_total": mkt_total_12m.shift(1)}),
              on="DATE", how="left")
ls.loc[ls["mkt_12m_total"] < 0, "lev"] = ls.loc[ls["mkt_12m_total"] < 0, "lev"] * bad_mkt_mult

# Defensive return
ls["ret_def"] = ls["lev"] * ls["ret_raw"]
ls = ls.dropna(subset=["ret_raw"]).reset_index(drop=True)
```

```
### Plots
def wealth(x): return (1.0 + x).cumprod()
ls_plot = ls[ls["DATE"] >= "1964-06-01"]
plt.figure()
plt.plot(ls_plot["DATE"], wealth(ls_plot["ret_raw"]))
plt.title("Top20/Bottom20 (12-2) - RAW (no defensive overlay)")
plt.xlabel("Date"); plt.ylabel("Portfolio value (start = $1)")
plt.tight_layout(); plt.savefig("plot_wealth_raw.png", dpi=150)

plt.figure()
plt.plot(ls_plot["DATE"], wealth(ls_plot["ret_def"]))
plt.title("Top20/Bottom20 (12-2) - DEFENSIVE overlay")
plt.xlabel("Date"); plt.ylabel("Portfolio value (start = $1)")
plt.tight_layout(); plt.savefig("plot_wealth_defensive.png", dpi=150)
```

```
### Regressions
# Regression frame
reg = ls.drop(columns=[c for c in ["Mkt-RF","RF"] if c in ls.columns]) \
        .merge(factors, on="DATE", how="left")

# FF3
X3 = sm.add_constant(reg[["Mkt-RF","SMB","HML"]])
res3_def = sm.OLS(reg["ret_def"], X3, missing="drop").fit()

# FF5
X5 = sm.add_constant(reg[["Mkt-RF","SMB","HML","RMW","CMA"]])
res5_def = sm.OLS(reg["ret_def"], X5, missing="drop").fit()

# Alpha comparison
def get_alpha(df_with_returns, y_col, factors_df, model="FF5"):
    merged = df_with_returns[["DATE", y_col]].merge(factors_df, on="DATE", how="left")
    if model == "FF3":
        Xcols = ["Mkt-RF", "SMB", "HML"]
    else:
        Xcols = ["Mkt-RF", "SMB", "HML", "RMW", "CMA"]
    X = sm.add_constant(merged[Xcols])
    res = sm.OLS(merged[y_col], X, missing="drop").fit()
    a_m = res.params["const"]
    a_y = (1 + a_m)**12 - 1
    return res, a_m, a_y

res3_raw,  a3m_raw,  a3y_raw  = get_alpha(ls, "ret_raw", factors, "FF3")
res3_def2, a3m_def,  a3y_def  = get_alpha(ls, "ret_def", factors, "FF3")
res5_raw,  a5m_raw,  a5y_raw  = get_alpha(ls, "ret_raw", factors, "FF5")
res5_def2, a5m_def,  a5y_def  = get_alpha(ls, "ret_def", factors, "FF5")
```

```
### Prints
print("\n================ ALPHA COMPARISON (monthly / annualized) ================\n")
print(f"FF3  Raw:        {a3m_raw*100:.3f}%  |  {a3y_raw*100:.2f}%")
print(f"FF3  Defensive:  {a3m_def*100:.3f}%  |  {a3y_def*100:.2f}%")
print(f"FF5  Raw:        {a5m_raw*100:.3f}%  |  {a5y_raw*100:.2f}%")
print(f"FF5  Defensive:  {a5m_def*100:.3f}%  |  {a5y_def*100:.2f}%")

print("\n================ FF3 Raw Strategy ================\n")
print(res3_raw.summary())
print("\n================ FF3 Defensive Strategy ================\n")
print(res3_def2.summary())
print("\n================ FF5 Raw Strategy ================\n")
print(res5_raw.summary())
print("\n================ FF5 Defensive Strategy ================\n")
print(res5_def2.summary())
```